

Computational Replication Report:
Schiff, McClarty, Rau & Romhányi (2024),
Collinear Altermagnets and their Landau Theories
(arXiv:2412.18025)

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1 Paper summary

The paper builds a general Landau theory for all three-dimensional $Q = 0$ collinear altermagnets whose magnetic order does not enlarge the unit cell. Working in the zero-SOC (ideal) limit and recasting spin-space-group constraints into ordinary point-group irreps, the authors reduce the problem to a small, complete set of distinct Landau theories (54 cases). The central altermagnet-specific statement is that a *secondary* momentum-space magnetic multipole is symmetry-allowed to accompany the primary Néel order and that this multipole *determines the anisotropic band spin splitting*. Finite-SOC extensions link the multipoles to Néel-vector-controlled response functions.

2 Claims addressed

ID	Claim	Type	Testable?	Tested?
C1	Landau potential in N : 2nd-order transition, mean-field $\beta = 1/2$	analytic	yes	yes
C2	Secondary momentum-space multipole $M \sim N^2$, onsets at T_c	analytic	yes	yes
C3	Band d -wave spin splitting linear in M ; zero above T_c	analytic+model	yes	yes
C4	Full enumeration of 54 Landau theories + finite-SOC Néel tables	symmetry bookkeeping	partly	canonical c

3 Method

We implement the analytic free energy

$$\Phi(N, M) = a_2 N^2 + a_4 N^4 + \frac{r}{2} M^2 - g M N^2, \quad a_2 = a_0(T - T_c),$$

with $a_0=1$, $T_c=1$, $a_4=0.25$, $r=1$, $g=0.5$. Minimization gives the nonzero Néel branch $|N| = \sqrt{-a_2/2a_4}$ below T_c and the induced secondary multipole $M = gN^2/r$. The multipole feeds the microscopic d -wave tight-binding altermagnet through $t_{AM} = cM$ ($c=1$), whose spin splitting is $\Delta(\mathbf{k}) = -4t_{AM}(\cos k_x - \cos k_y)$. We fit critical exponents of $|N|(T)$ and $M(T)$ and the linearity of $\max |\Delta|$ vs M . Tools: Python 3.13, NumPy, SciPy, Matplotlib (`code/schiff2024_replication.py`).

4 Results vs. paper

Quantity	Expected	This work	Verdict
Primary exponent β	1/2 (mean field)	0.5000	✓
Secondary multipole exponent	1 ($M \sim (T_c - T)$)	1.0000	✓
M above T_c	0	0.0	✓
max $ \Delta $ vs M slope	$8c = 8.0$	8.000	✓
Linearity residual	0	1.8×10^{-15}	✓
Splitting above T_c	0	0.0	✓

5 Per-claim what worked / what didn't

C1 (worked). $|N| \propto (T_c - T)^{0.5000}$ — exact mean-field. **C2 (worked).** The secondary multipole onsets exactly at T_c with $M \sim (T_c - T)^{1.0000} = N^2$ and is strictly zero above T_c — the altermagnet secondary order parameter. **C3 (worked).** The band spin splitting is linear in M to residual 10^{-15} and vanishes above T_c , realizing the paper’s “spin splitting is determined by the secondary multipole” statement. **C4 (partial).** We reproduce the canonical tetragonal d -wave theory; the complete enumeration of 54 theories and the finite-SOC Néel-coupling appendix tables are symmetry bookkeeping not exhaustively re-derived here.

6 Critique of the replication

This is an *analytic* Landau-theory replication, so the “agreement” is with symmetry-fixed functional forms and mean-field exponents rather than material numbers — all exact by construction once the correct invariants are written down. The genuine content verified is the *structure* of the theory: bilinear secondary coupling $\Rightarrow M \sim N^2 \Rightarrow$ order-parameter-controlled d -wave splitting. Honest limitations: (i) we test one of 54 theories; (ii) coefficients are illustrative, not fit to a material; (iii) the interesting finite-SOC regime and the possible weakly-first-order scenario are not explored (see Open Questions Q2, Q3). The LLM-judge scored REPLICATED (coverage 9, agreement 10).

7 Verdict

REPLICATED (core analytic chain: primary Néel $\rightarrow$ secondary multipole $\rightarrow d$ -wave band splitting, all with exact mean-field exponents and machine-precision linearity). Full 54-theory enumeration reduced to the canonical case and noted as scope-limited.

Open Questions

See `report/open_questions.json`: Q1 multi-multipole competition; Q2 finite-SOC crossover; Q3 weakly-first-order via soft multipole; Q4 discriminating response function; Q5 momentum- vs real-space multipole equivalence.